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Profitability of a project

Financial UDF pack

Part 1: How it works, financial explanations

Patou Tips #60

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Profitability of a project
Financial UDF pack (1/3)

About this financial package...

This package of 8 DAX functions is designed to provide access to key performance indicators (KPIs) representative of project profitability, such as IRR, NPV, and payback. It also allows for the calculation of the metrics such as DF, FCF, and DCF.

In order to easily transpose these functions into your projects, and to use these functions as DAX functions, these 8 functions are developed in UDF (User Defined Function) mode.

Functions	What it is?
DF.PatouTips	Calculate DF (Discounted Factor) by period
FCF.PatouTips	Calculate FCF (Discounted Free Cash Flow) by period
DCF.PatouTips	Calculate DCF (Discounted Cash Flow) by period
IRR.PatouTips	Calculate IRR (Internal Rate Return) by period
NPV.PatouTips	Calculate NPV (Net Present Value) by period
NPV.Total.PatouTips	Calculate NPV (Net Present Value) at end of period
Payback.Year.PatouTips	Calculate the year of the project payback
Payback.Nb.Year.PatouTips	Calculate the number of year of the project payback

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Profitability of a project
Financial UDF pack (2/3)

About this financial package...

💡 Functions are created to be used independently, in whole or in part. Functions are not linked to each other by code. Each function generates an value for each period used to calculate the profitability of a project. These values can be displayed as a matrix or a graph for example.

Project period

2024 2028

Metrics

990 000
Revenue

-600 000
Cost

-260 000
Investment

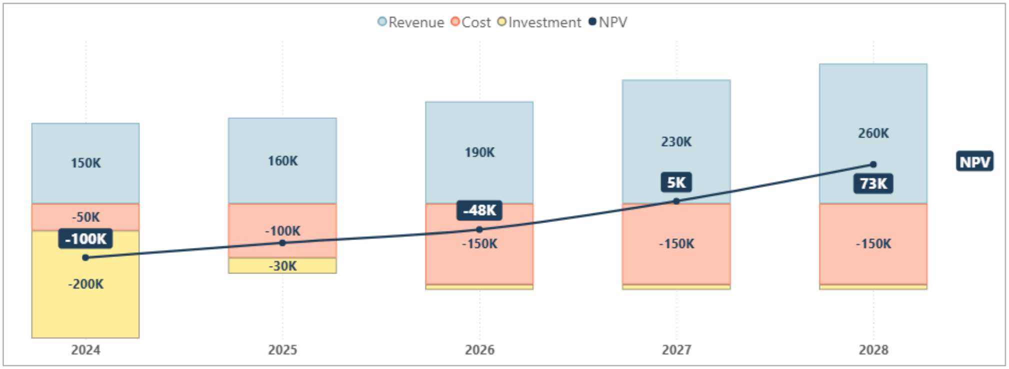
KPI

72 959
NPV Total

33,6 %
IRR

2027
Payback Year

4
Payback Nb Year



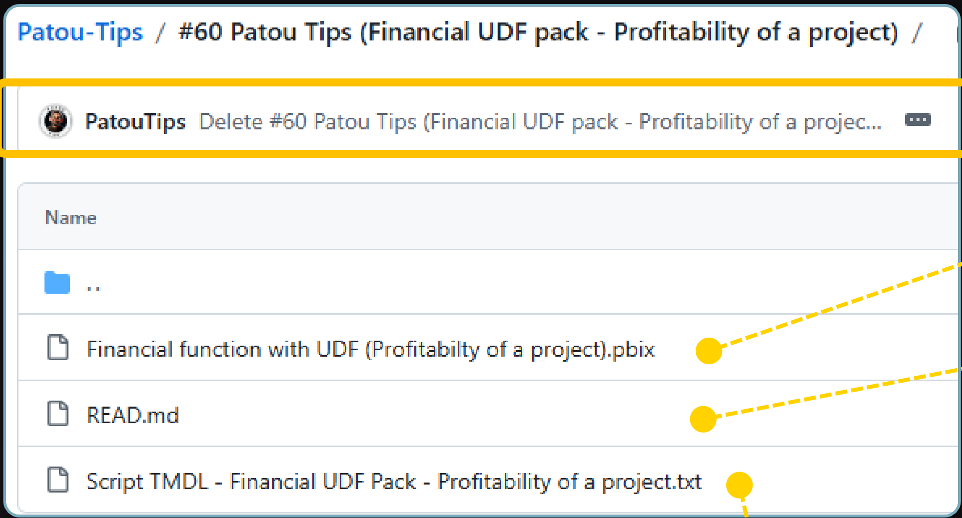
Year	Revenue	Cost	Investment	FCF	DF	DCF	NPV
2024	150 000	-50 000	-200 000	-100 000	1,00	-100 000	-100 000
2025	160 000	-100 000	-30 000	30 000	0,91	27 273	-72 727
2026	190 000	-150 000	-10 000	30 000	0,83	24 793	-47 934
2027	230 000	-150 000	-10 000	70 000	0,75	52 592	4 658
2028	260 000	-150 000	-10 000	100 000	0,68	68 301	72 959
Total	**990 000**	**-600 000**	**-260 000**	**130 000**			

0 Profitability of a project
Financial UDF pack (3/3)

About this financial package...

Find all the resources on the Patou Tips GitHub

https://github.com/PatouTips/Patou-Tips/



Repository

PowerBI file example

To go further...

Script TMDL to transpose functions in your PowerBI project

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Finance UDF (User Defined Function) Pack

Profitability of a project

Part 1:
How it works?
Financial explanations

Profitability of a project
How it works? (1/4)

2 ways* to download UDF functions from the Patou Tips GitHub repository, chapter #60:

1 - Open the file "Script TMDL - Financial UDF Pack - Profitability of a project.txt", copy the contents of this text file, then paste the entire script into the TMDL view of your Power BI project (see Patou Tips #58 for more information).

Script TMDL - Financial UDF Pack - Profitability of a project.txt

2 - Go to the "READ.md" file and click the copy icon in chapter "3 - Code".

3 - Code

```
createOrReplace

    /// Discounted Cash Flow (DCF) is a valuation method that estimates the value
    function 'DCF.Patou.Tips' = ```
```

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Profitability of a project
How it works? (2/4)

✏️ Create a measure that calls one of the "Patou.Tips" functions, for example, write the word "Patou".

The function-specific instructions are also summarized in the pop-up window that appears when the measure is inserted.

```
NPV = NPV.Patou.Tips([Revenue]
```

```
NPV.Patou.Tips(Revenue_measure: expr, Cost_measure: expr, Investment_measure: expr, Year_date_of_calendar_table: anyref, Discounted_rate: decimal)
```

Net Present Value (NPV). This is the cumalative Discounted Cash Flow (DCF), for each period of the project. NPV evaluates the profitability of an investment by comparing the present value of expected future cash flows to the initial investment. Put Costs and Investment in negative value. The Discounted Rate is a decimal value representing a percentage. For example, 0.101 for 10.1%.

1

Profitability of a project
How it works? (3/4)

Download the Power BI file which will give you a complete example of its use in a financial project.

Financial function with UDF (Profitabilty of a project).pbix

The dataset uses in this Patou Tips comes from the fictitious Parisian company "IceCream' Macaron," taken from my book "Business Intelligence".

Year	Revenue	Cost	Investment
2024	150 000 €	-50 000 €	-200 000 €
2025	160 000 €	-100 000 €	-30 000 €
2026	190 000 €	-150 000 €	-10 000 €
2027	230 000 €	-150 000 €	-10 000 €
2028	260 000 €	-150 000 €	-10 000 €
2029	275 000 €	-150 000 €	-10 000 €
2030	300 000 €	-150 000 €	-10 000 €
2031	330 000 €	-150 000 €	-10 000 €
2032	360 000 €	-150 000 €	-10 000 €

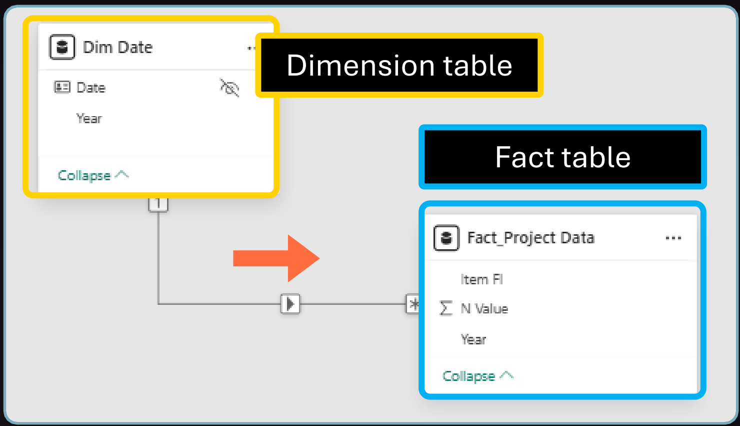
→ This project represents the launch of a new store for the company "IceCream' Macaron." It involves a significant initial investment and residual investments for possible improvements to the store structure.

1 Profitability of a project
How it works? (4/4)

To use the UDF of this financial pack, you need at least 3 measures :

- ✓ Revenue,
- ✓ Cost,
- ✓ Investment
- ✓ And a calendar table with the "Year" field at least

Note: Cost and Investment measures be in negative values.



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Profitability of a project
Financial explanations (1/3)

Project inputs

Revenue + Costs + Investment =

Discounted Rate*

*Discounted Rate: WACC, inflation rate, internal financial rate of return…

Metrics

Free Cash Flow (FCF)

Significant financial indicators must be taken into account

Discount Factor (DF)

Depreciation of the "Discounted Rate" for each project term

Discounted Cash Flow (DCF)

= Free Cash Flow x Discount Rate

KPI

IRR

Payback Period

NPV

NPV = Cumulative DCF

Profitability of a project is good when these 3 conditions (KPI) are met…

- IRR > Profitability rate required
- Payback period < Project Term
- NPV > 0

1 Profitability of a project
Financial explanations (2/3)

The profitability of a project is essential for making informed investment decisions. This assessment, which incorporates the time value of money, the **discount factor (DF)**, involves using tools such as:

The Net Present Value (NPV) calculates the difference between the present value of future cash flows and the initial investment. A positive NPV indicates that the project generates more value than its cost, thus signaling a profitable opportunity. Conversely, a negative NPV suggests an unviable project.

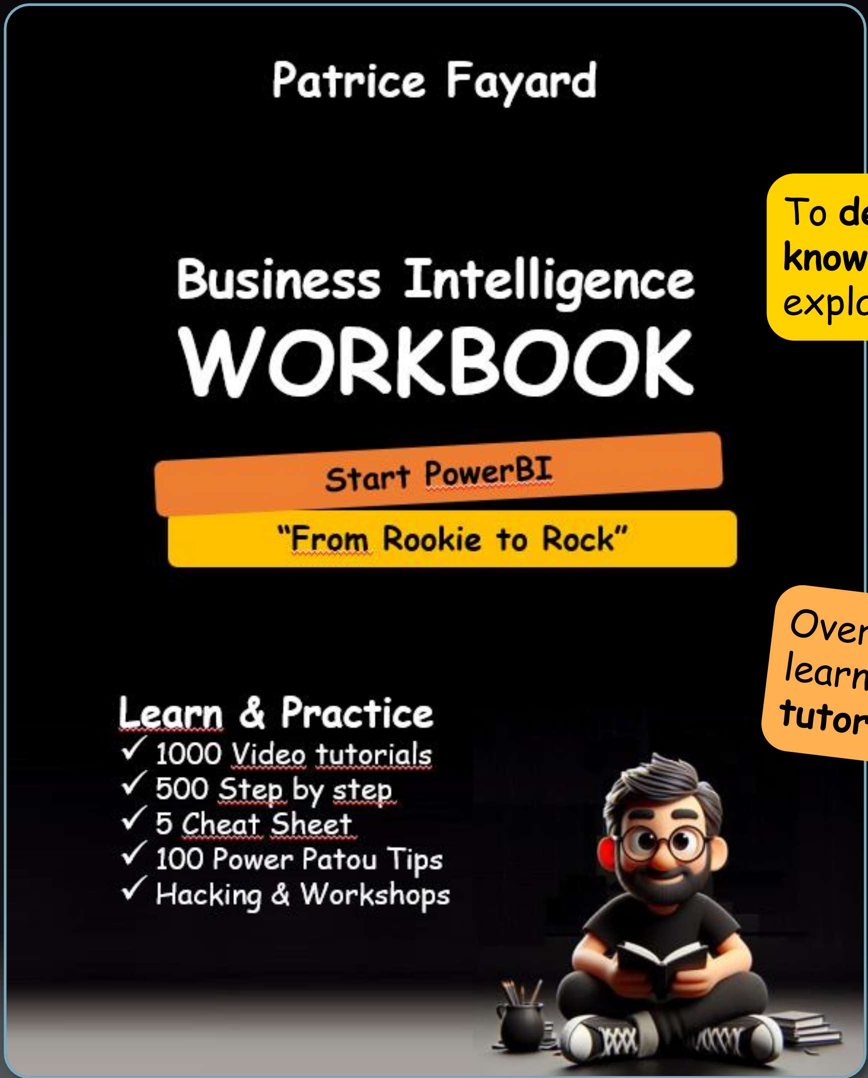
The Internal Rate of Return (IRR) represents the discount factor (DF) that makes the NPV equal to zero. It allows you to compare the project's profitability to a minimum required rate of return (cost of capital). An IRR higher than this rate means the project is attractive.

1 Profitability of a project
Financial explanations (3/3)

An indicator such as the payback period complements this analysis by assessing the time required to obtain a return on investment.

In summary, these tools offer a clear view of risks and opportunities, helping investors and managers to prioritize the most profitable projects, in line with their strategic objectives.

Coming soon, in 2026 / 2027!



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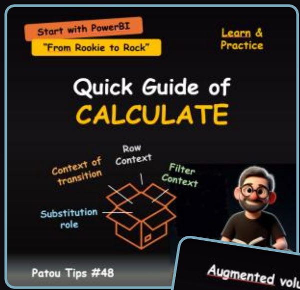
Project version shown

Learn and practice

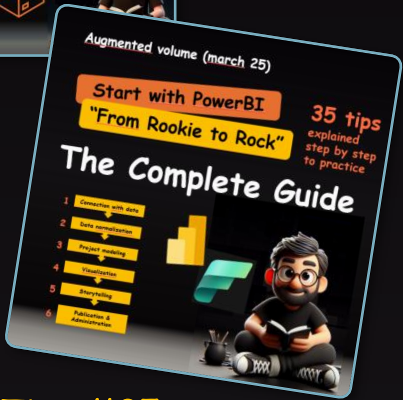
Find past issues of "Patou Tips" and download resources to practice on GitHub

Patou Tips #22
PowerBI projects with Teams

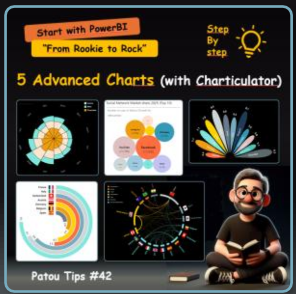


Patou Tips #48
Quick Guide of CALCULATE

Patou Tips #23
Quick Guide Visualization & Storytelling



Patou Tips #25
Start with PowerBI: The complete Guide



Patou Tips #42
5 Advanced Charts (with Charticulator)

Resources on GitHub
https://github.com/PatouTips/Patou-Tips

Patou Tips

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